

EFFICIENT BAYESIAN INFERENCE FROM NOISY PAIRWISE COMPARISONS

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Anonymous submission

ABSTRACT

Evaluating generative models is challenging because standard metrics often fail to reflect human preferences. Human evaluations are more reliable but costly and noisy, as participants vary in expertise, attention, and diligence. Pairwise comparisons improve consistency, yet aggregating them into overall quality scores requires careful modeling. Bradley-Terry-based methods update item scores from comparisons, but existing approaches either ignore rater variability or lack convergence guarantees, limiting robustness and interpretability. We introduce BBQ, a Bayesian Bradley-Terry variant that explicitly models rater quality, downweighting or removing unreliable participants, and provides guaranteed monotonic likelihood convergence through an Expectation-Maximization algorithm. Across eight human-evaluation datasets and against two strong baselines (Crowd-BT and Bayes-BT), BBQ is the method that most frequently identifies the top-ranked item and recovers the best overall ranking on five of the eight datasets, placing second on the remaining three. The gap is largest exactly where it matters most: on noisy, unscreened crowdsourced comparisons, BBQ reaches 61.92% top-1 agreement versus 33.15% (Crowd-BT) and 24.32% (Bayes-BT), while still converging in seconds and producing well-calibrated uncertainty estimates. Together, these results show that explicitly modeling rater quality yields more robust and interpretable rankings than baseline Bradley-Terry models, especially with noisy or crowdsourced raters, enabling more reliable and cost-effective human evaluation of generative models.

1 INTRODUCTION

Evaluating generative models is challenging, particularly for large language models (LLMs) and image generators, where standard metrics often fail to reflect human preferences. Metrics such as BLEU (Papineni et al., 2002) and perplexity (Jelinek, 1998) for LLMs, or PSNR (Gonzalez, 2009), MS-SSIM (Wang et al., 2003), and FID (Heusel et al., 2017) for image models, provide only limited insight into perceived quality (Mentzer et al., 2020; CLIC, 2025; Chiang et al., 2024). As a result, human evaluations remain indispensable for establishing meaningful rankings between models. Accordingly, leaderboards and comparative studies of large language models and Learned Image Compression (LIC) methods place strong emphasis on human preference data.

However, human evaluations are expensive and time-consuming, making it crucial to design protocols that are efficient while minimizing subjectivity and noise. In this context, pairwise comparisons, in which participants choose between two items rather than providing absolute scores, represent an effective and practical form of human evaluation (Zerman et al., 2018; Wang et al., 2023): they are generally easier for participants and produce more consistent judgments. Collecting all pairwise comparisons is infeasible, however, because the number of pairs grows quadratically with the number of items. When only a limited set of comparisons is available, simple statistics such as win rates are not sufficient to derive overall quality scores, because a win rate depends on the quality of the items it is compared against. This gap between the comparisons one can afford to collect and the reliable rankings one needs is precisely what a good aggregation method must close.

To produce an overall ranking that can be displayed on leaderboards, pairwise comparisons must be aggregated effectively. This motivates the development of robust and efficient aggregation methods. The Bradley-Terry (BT) (Bradley & Terry, 1952) model addresses this problem by iteratively updat-

ing item scores based on comparison outcomes. It has become a standard approach for aggregating pairwise judgments across domains.

Despite its usefulness, the standard Bradley–Terry model has several limitations in practice. There are multiple algorithms for estimating BT parameters, including iterative scaling methods (Ford Jr, 1957), gradient descent, and the minorization maximization (MM) algorithm (Hunter, 2004). The MM algorithm guarantees convergence to the global maximum for the basic BT model, whereas gradient-based methods, which are closely related to the Elo rating system, are widely used in practice but provide no such guarantees (Hunter, 2004). For extended BT models that incorporate factors such as home field advantage (Agresti, 2010), multiple comparisons (Plackett, 1975; Luce et al., 1959), or ties (Rao & Kupper, 1967), even MM algorithms may converge only to local optima (Hunter, 2004). A further challenge is that human evaluations are inherently noisy: participants may rush, guess, or lose focus, leading to uneven reliability across raters. If such variability is ignored, unreliable raters can distort item scores and reduce ranking stability.

We propose a Bayesian Bradley–Terry variant that jointly models item quality and rater reliability. To the best of our knowledge, no prior work combines the BT framework with a Bayesian formulation that explicitly models rater quality while providing closed-form EM updates. This distinction is empirically consequential rather than cosmetic: the previous approaches that account for noisy raters rely on gradient-based optimization without convergence guarantees, and, as our experiments show, this leads them to fail outright with very few raters and to degrade sharply on noisy crowd-sourced data, precisely the regime where our EM-based method converges reliably and retains its accuracy. The Bayesian framework addresses epistemic uncertainty and provides regularization. To capture variability in participant behavior, we introduce rater-specific parameters that reflect how consistent or trustworthy each participant is, allowing the model to adjust the influence of individual comparisons. We derive an EM algorithm that efficiently estimates item and rater parameters via a latent-variable formulation, guaranteeing monotonic likelihood improvement. This approach allows partially reliable raters to contribute meaningful information while reducing the impact of inattentive or inconsistent participants. By modeling rater quality and adopting a Bayesian framework, our method also provides uncertainty estimates for item scores, facilitating more interpretable rankings and robust comparisons across studies.

We demonstrate the effectiveness of our approach on eight human evaluation datasets spanning language and image domains, and introduce a newly collected *inhomogeneous rater quality (IHQ)* dataset¹. Directly comparing BBQ against the two strongest available baselines, Crowd-BT and Bayes-BT, we find that BBQ identifies the top-ranked item most often and recovers the most accurate overall ranking on the majority of datasets, while converging in seconds rather than minutes. The advantage is most pronounced precisely where existing methods struggle: on noisy, unscreened crowdsourced comparisons, BBQ nearly doubles the top-1 agreement of the strongest baseline, and it does so without sacrificing accuracy on clean, expert-labeled data. Beyond generative model evaluation, the proposed framework is applicable to a wide range of scenarios where noisy pairwise comparisons must be transformed into reliable global rankings. Overall, these results position BBQ as a practical route to more cost-efficient, interpretable, and reproducible human studies for evaluating AI-generated content.

2 RELATED WORK

Human studies often require ranking items. Presenting participants with a choice between two items rather than a single item with a score increases sensitivity to subtle differences and reduces variability in responses (Zerman et al., 2018; Wang et al., 2023). Consequently, many leaderboards for machine learning models rely on pairwise comparisons. In such setups, participants indicate their preference between two alternatives, and these preferences are then aggregated to produce overall rankings. Notable examples include the Chatbot Arena (Chiang et al., 2024) and the CLIC image compression challenge (CLIC, 2025), which use pairwise comparisons and combine them using variants of the Bradley–Terry (BT) model.

The Bradley–Terry model (Bradley & Terry, 1952) was originally developed to rank competitors in sports. It provides a probabilistic framework to estimate the likelihood that one item is preferred

¹The IHQ dataset will be released upon acceptance.

over another. The model converts pairwise comparisons into a ranking, making it a cornerstone in studies across games, consumer preferences, and other applications. Several extensions have been proposed to broaden its applicability. For instance, the Plackett-Luce model generalizes the BT framework from pairwise comparisons to rankings over multiple items (Plackett, 1975; Luce et al., 1959), defining a probability distribution over permutations by multiplying successive BT probabilities (Luce et al., 1959). Other modifications address specific contexts, such as modeling home-field advantages (Agresti, 2010), incorporating ties (Rao & Kupper, 1967), or handling comparisons between groups instead of individuals (Huang et al., 2006).

Maximum likelihood estimation (MLE) is commonly used to infer the parameters of the basic BT model. Zermelo (1929) introduced an iterative approach to compute these estimates, which has become widely adopted. Later, Lange et al. (2000) demonstrated that this procedure is a particular case of minorization-maximization (MM) algorithms, which iteratively optimize surrogate functions to reach a local maximum of the likelihood. Hunter (2004) extended MM algorithms to generalized BT models and established conditions guaranteeing convergence to the MLE. For the classical BT model, the optimization is convex, ensuring convergence to the global optimum (Hunter, 2004). However, for extensions such as home-field advantage, multiple comparisons, or ties, MM algorithms may converge only to a local maximum (Hunter, 2004).

Bayesian formulations of the BT model have been explored to incorporate prior knowledge and regularization (Adams, 2005; Guiver & Snelson, 2009; Caron & Doucet, 2012). Caron & Doucet (2012) showed that MM algorithms can be interpreted as Expectation-Maximization (EM) procedures, enabling Bayesian inference via Gibbs sampling. This formulation provides tractable complete-data likelihoods and ensures convergence of the resulting Markov Chain Monte Carlo methods. The Bayesian perspective smooths posterior estimates, reducing susceptibility to local maxima, and allows for uncertainty quantification in the estimated item skills.

Annotator quality models represent a critical extension addressing the assumption that all comparisons are equally reliable. In crowdsourced evaluations, participant quality can vary widely, motivating approaches that explicitly account for annotator reliability (Chen et al., 2013). Chen et al. (2013) proposed a Bayesian model, Crowd-BT, that jointly estimates both item quality and rater reliability using annotator-specific parameters. Their approach places Gaussian priors on item and rater parameters, and incorporates scaling factors in the likelihood to modulate individual annotator influence. Crucially, they perform posterior inference using gradient-based optimization, which offers no convergence guarantees and can be difficult in high-dimensional spaces due to potential local optima; our experiments confirm this concretely, as Crowd-BT fails to converge with only one or two raters and slows to roughly 15 minutes per bootstrap iteration on the largest dataset. In direct contrast, our proposed EM-based method provides guaranteed monotonic likelihood improvement and, as we report in Section 4, converges within seconds on every dataset while remaining accurate under noise. Crowd-BT is the same rater model that Google Research implements in a simplified Elo-based form, which is widely used in the Challenge on Learned Image Compression (CLIC) (CLIC, 2025) and related research (Mentzer et al., 2020; Ballé et al., 2025), making it the most relevant and competitive baseline for our comparisons.

To the best of our knowledge, no prior work combines the Bradley-Terry model with a Bayesian formulation that explicitly models rater quality while also providing closed-form EM updates. This positions BBQ against two clearly delineated points of comparison. Relative to Crowd-BT (Chen et al., 2013), which uses Gaussian priors and gradient-based optimization without convergence guarantees, our approach leverages conjugate priors and an EM algorithm that ensures monotonic likelihood improvement; the practical payoff is faster and more reliable convergence together with markedly higher top-1 agreement on noisy data. Relative to the simplified Elo-based rater heuristic widely used in CLIC (CLIC, 2025), our method provides a principled Bayesian treatment that additionally yields calibrated uncertainty estimates and interpretable rater-quality parameters, neither of which the heuristic offers.

3 METHODOLOGY

3.1 BRADLEY-TERRY MODEL WITH RATER QUALITY

The objective of converting a set of noisy pairwise comparisons into a reliable ranking of items is a fundamental problem in machine learning and statistics. Consider a set of K items that are repeatedly compared with one another in pairs by a set of R raters. The data, which we denote as D , consists of the outcomes of these comparisons. For two items i and j of this set, Bradley & Terry (1952) suggested the following model:

$$P(i \text{ beats } j) = \frac{\lambda_i}{\lambda_i + \lambda_j} \quad (1)$$

where $\lambda_k > 0$ is a parameter associated with item $k \in \{1, 2, \dots, K\}$ that represents its skill rating. This model provides a clear and interpretable way to infer item scores from a set of observed wins and losses. However, it operates under the simplifying assumption that all comparisons are equally reliable, an assumption that is often violated in human evaluation, where participants vary in expertise, attention, and diligence and thus produce judgments of uneven reliability. The empirical cost of this assumption is visible later in Table 1: on the unscreened crowdsourced subset, ignoring rater variability drags top-1 agreement down to the 24–33% range, motivating the rater-aware model we develop next.

To account for varying rater reliability, we introduce a rater-specific quality parameter $q_r \in [0, 1]$. Intuitively, with probability q_r , rater r makes an informed judgment following the Bradley-Terry model. With probability $1 - q_r$, the rater guesses randomly, as if flipping a fair coin between the two items. This leads to the following mixture model for the probability that rater r ranks item i above item j :

$$P(r \text{ ranks } i \text{ above } j) = q_r \left(\frac{\lambda_i}{\lambda_i + \lambda_j} \right) + (1 - q_r) \left(\frac{1}{2} \right) \quad (2)$$

The log-likelihood of the data is given by:

$$\begin{aligned} \log P(D \mid \lambda, q) = & \sum_{r=1}^R \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K \left[w_{r,ij} \log \left(q_r \frac{\lambda_i}{\lambda_i + \lambda_j} + (1 - q_r) \frac{1}{2} \right) \right. \\ & \left. + (n_{r,ij} - w_{r,ij}) \log \left(q_r \frac{\lambda_j}{\lambda_i + \lambda_j} + (1 - q_r) \frac{1}{2} \right) \right], \end{aligned} \quad (3)$$

where $n_{r,ij}$ denotes the total number of comparisons between items i and j by rater r , and $w_{r,ij}$ is the number of times rater r ranks item i above item j . There is no closed-form maximum likelihood estimator for this likelihood function, so the optimal parameters λ and q cannot be derived analytically. The choice of optimizer therefore matters in practice. A generic iterative method such as gradient descent, the route taken by the Crowd-BT baseline, requires learning-rate tuning and offers no guarantee of convergence, a weakness we later observe directly when it fails to converge in the sparse-rater regime. The Expectation–Maximization (EM) algorithm we adopt instead avoids learning-rate tuning altogether and guarantees a monotonic increase of the observed-data likelihood and convergence to a stationary point (Dempster et al., 1977), and the resulting closed-form updates are what let BBQ converge within seconds across all eight datasets in Section 4.

3.2 THURSTONIAN INTERPRETATION

Following Caron & Doucet (2012), we interpret the Bradley-Terry model under a Thurstonian framework (Diaconis, 1988). In this perspective, a comparison between items i and j is conceptualized as a race, where each item has a random arrival time, Y_i and Y_j , respectively. These arrival times are assumed to follow exponential distributions:

$$Y_i \sim \mathcal{E}(\lambda_i), \quad Y_j \sim \mathcal{E}(\lambda_j), \quad (4)$$

and the item with the smaller arrival time is declared the winner. This leads directly to the standard Bradley-Terry probability:

$$P(i \text{ beats } j) = P(Y_i < Y_j) = \frac{\lambda_i}{\lambda_i + \lambda_j}. \quad (5)$$

For the EM algorithm, we introduce latent variables to simplify the complete-data likelihood. First, we define an indicator variable

$$A_{r,ij}^{(c)} \sim \text{Bernoulli}(q_r), \quad (6)$$

which denotes whether the c -th comparison of items i and j by rater r follows the Bradley-Terry model. Using these indicators, we define the latent variable $Z_{r,ij}$ as the sum of the minimal arrival times across the $n_{r,ij}$ comparisons by rater r :

$$Z_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)} \min(Y_i^{(c)}, Y_j^{(c)}). \quad (7)$$

Conditioned on $m_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)}$, the variable $Z_{r,ij}$ follows a Gamma distribution,

$$Z_{r,ij} \mid m_{r,ij} \sim \Gamma(m_{r,ij}, \lambda_i + \lambda_j), \quad (8)$$

where $\Gamma(\alpha, \beta)$ denotes the Gamma distribution with shape parameter α and inverse scale β . This Gamma-distributed latent variable formulation allows for a tractable EM update while accounting for rater-specific quality.

3.3 EXPECTATION-MAXIMIZATION UPDATES

The EM algorithm is an iterative method for finding maximum a posteriori (MAP) estimates for our model parameters, λ and q , by treating the rater's quality and the unobserved arrival times from the Thurstonian interpretation as latent variables. The algorithm is guaranteed to converge to a stationary point of the posterior distribution.

First, we specify prior distributions for the parameters. The item skills λ_k are assigned a Gamma prior, $\lambda_k \sim \Gamma(a, b)$, which is a conjugate prior for the exponential distribution. Each rater's quality parameter, q_r , is given a Beta prior, $q_r \sim B(\alpha, \beta)$.

The core of the EM algorithm is the iterative maximization of the expected complete-data log-posterior, conditioned on the current parameter estimates (λ^*, q^*) . The algorithm proceeds by alternating between two steps until convergence:

E-step: Expectation In the E-step, we compute the expected complete-data log-posterior, a function we denote as Q . This function represents the expected value of the log-posterior of all observed and latent variables, given our observed data and the current parameter estimates from the previous iteration. It is defined as:

$$Q(\lambda, q \mid \lambda^*, q^*) = \mathbb{E}_{A,Z \mid D, \lambda^*, q^*} [\ell_c(\lambda, q; D, Z, A) + \log P(\lambda) + \log P(q)]. \quad (9)$$

The complete-data log-likelihood, ℓ_c , is further broken down into three components.

$$\ell_c(\lambda, q; D, Z, A) = \log P(Z \mid D, A, \lambda, q) + \log P(A \mid D, \lambda, q) + \log P(D \mid \lambda, q). \quad (10)$$

M-step: Maximization This step updates the model parameters by maximizing the Q function. By leveraging the expected values from the E-step, the M-step transforms the original complex optimization problem into simpler, closed-form updates. The key quantity that is used in both updates is the posterior probability that a given comparison from rater r follows the Bradley-Terry model. This quantity, denoted as γ , represents the weight of a rater's judgment based on how much it aligns with the model's current predictions. It is given by:

$$\gamma_{r,ij}^{(t-1)} = \frac{q_r^{(t-1)} y_{ij}^{(t-1)}}{q_r^{(t-1)} y_{ij}^{(t-1)} + (1 - q_r^{(t-1)}) \frac{1}{2}}, \quad (11)$$

where $y_{ij}^{(t-1)} = \frac{\lambda_i^{(t-1)}}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}}$ is the Bradley-Terry probability that item i beats item j . This posterior probability γ represents our confidence that a comparison was meaningful rather than random, given the current parameter estimates. Higher γ values indicate more trustworthy comparisons.

The new estimate for a rater’s quality, q_r , is calculated as a weighted average. The numerator sums up the ”effective number of wins” for that rater, where each win is weighted by the posterior probability (γ) that it was a meaningful, non-random judgment. This is combined with the hyperparameters from the Beta prior to regularizing the estimate. The denominator normalizes this sum by the total number of comparisons and prior parameters. This update intuitively increases a rater’s quality score if their judgments frequently align with the model’s predictions. The update is given by:

$$q_r^{(t)} = \frac{\sum_{i=1}^K \sum_{j=i+1}^K [w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)}] + (\alpha - 1)}{n_r + \alpha + \beta - 2}, \quad (12)$$

where n_r is the total number of comparisons by rater r .

The new estimate for an item’s skill, λ_i , is a ratio that balances two key quantities. The numerator is a sum of the ”effective wins” for item i across all raters, where each win is again weighted by the rater’s quality (γ). This term essentially represents the total positive evidence for item i . The denominator, on the other hand, accounts for the total comparisons item i was involved in, and acts as a normalizing factor. These terms are also regularized by the Gamma prior hyperparameters. The update is given by:

$$\lambda_i^{(t)} = \frac{\sum_{r=1}^R [\sum_{j=1, j \neq i}^K w_{r,ij} \gamma_{r,ij}^{(t-1)}] + (a - 1)}{\sum_{j=1, j \neq i}^K \left[\frac{\sum_{r=1}^R [w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)}]}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}} \right] + b}. \quad (13)$$

The derivation of the Expectation Maximization algorithm is provided in Appendix B.

4 EXPERIMENTAL RESULTS

To evaluate our **Bayesian Bradley-Terry** model with rater **Quality (BBQ)**, we benchmark it head-to-head against two competitive baselines under identical conditions: Bayesian Bradley-Terry (Bayes-BT) (Caron & Doucet, 2012), which is Bayesian but does not model rater quality, and Crowd-BT (Chen et al., 2013), a gradient-descent BT model that does model rater quality but lacks convergence guarantees. These two baselines let us isolate the two ingredients of BBQ: the Bayes-BT comparison isolates the value of modeling rater quality, and the Crowd-BT comparison isolates the value of our EM optimizer over gradient descent. For Crowd-BT, we use the same Google Research implementation (Google, 2025) employed by CLIC (2025) and Ballé et al. (2025), so the baseline is the exact system used in practice; all hyperparameters are reported in Section C.

To make the comparison as broad as possible, we evaluate all three BT variants on eight datasets spanning two domains: human preference benchmarks for large language models (HUMAINE (Team, 2025), MT-Bench (Zheng et al., 2023)) and for image compression (WD (Ballé et al., 2025), HiFiC (Mentzer et al., 2020), ConHa (Aczel & Wattenhofer, 2024)), together with a newly collected inhomogeneous rater quality (IHQ) dataset from the CLIC2024 image test set (CLIC, 2025). To isolate the effect of rater quality under controlled conditions, we partition IHQ into a screened and an unscreened subset, which lets us contrast high-quality, low-quality, and mixed comparisons directly on otherwise identical data. Details of the IHQ dataset creation are given in Section E, and a summary of all datasets, including comparisons, raters, and items, is provided in Section D and Table 2.

Ground truth rankings are not available for real datasets. We first validated the models on simulated datasets, where all three methods recover the same ordering as the number of samples increases, confirming that they agree in the large-sample limit and that any differences we report arise from the finite, noisy regime. For crowd-sourced datasets, we approximate the ground truth by the ranking obtained on the whole dataset, and we sanity-check this approximation by inspecting the top-1 item of each ranking. This check already separates the methods: on seven of the eight datasets all three

models agree on the top-ranked item, but on the challenging IHQ-unscreened subset only BBQ correctly identifies the reference image as the highest-quality item, whereas both Crowd-BT and Bayes-BT miss it.

A reliable aggregation method should reproduce the same ordering if the study is repeated, so we quantify this stability directly through bootstrapping, which estimates the variability of the rankings under resampling; the procedure is detailed in Section F. Two datasets warrant a caveat when reading these stability numbers: the WD and HiFiC studies used active selection of comparison pairs (maximum information gain for WD, binary search for HiFiC), so their bootstrapping results should be interpreted with corresponding caution rather than as clean like-for-like comparisons.

We report two complementary metrics, both oriented so that higher is better. Top-1 agreement is the fraction of bootstrapped samples that identify the same best item as the full dataset, and it directly measures how reliably a method pins down the single best model. Kendall’s Tau (τ) measures ordinal correlation between rankings (Kendall, 1938) and captures the stability of the ranking as a whole. Together they let us distinguish the case where the top choice matters (top-1 agreement) from the case where the full ordering matters (Kendall’s τ), a distinction that turns out to be important for reading the results below.

4.1 PERFORMANCE ACROSS DATASETS

Table 1 summarizes the head-to-head comparison of the three methods on all eight datasets, along both metrics. As a general trend, datasets with more comparisons per model, such as MT-Bench, WD, and HiFiC, are easier for every method; HUMAINE is the informative exception, showing that rater consistency and diversity, not just raw comparison count, drive performance.

Read as a whole, the table tells a consistent story in BBQ’s favor. BBQ never loses on top-1 agreement: it attains the highest top-1 agreement on every one of the eight datasets, tying the baselines at 100% on the three easiest (MT-Bench, WD, HiFiC) and pulling clearly ahead on the rest. On Kendall’s τ it recovers the most accurate overall ranking on five of the eight datasets, and where it does not win it finishes a close second rather than trailing. No other method matches this combination of never being beaten on top-1 agreement and leading the majority of Kendall’s τ comparisons.

The only three datasets on which BBQ places second in Kendall’s τ (MT-Bench, WD, and IHQ-screened) are precisely the high-quality, homogeneous ones, and even there the margins are slight: on MT-Bench all models perfectly recover the top item, the WD study of Ballé et al. (2025) uses only five, likely expert, raters, and IHQ-screened raters were explicitly filtered for quality. When rater quality is uniformly high there is simply little rater variability left to model, so BBQ’s advantage narrows to a fraction of a τ point rather than reversing, and its top-1 agreement remains best-in-class (99.80% on IHQ-screened).

The IHQ subsets make the central pattern unmistakable, because they hold the data fixed and vary only rater quality. As noise increases from screened to unscreened raters, the gap in BBQ’s favor widens dramatically. On IHQ-screened all three methods are within about a percentage point on top-1 agreement (98.57, 98.90, and 99.80% for Crowd-BT, Bayes-BT, and BBQ), but on IHQ-unscreened BBQ reaches 61.92% while Crowd-BT and Bayes-BT collapse to 33.15% and 24.32%, so BBQ roughly doubles the better baseline. On the full mixed IHQ-all set the ordering is just as clear: BBQ attains 99.32% top-1 agreement against 85.31% (Crowd-BT) and 75.07% (Bayes-BT). Tellingly, Crowd-BT models rater quality yet still degrades sharply once screening is removed, most likely because each crowdsourced rater provides fewer than 40 comparisons, too few for its gradient-based estimator to pin down reliability; BBQ, by contrast, gives up only about half a percentage point of top-1 agreement between its screened and all conditions, evidence of genuine robustness to low-quality raters.

This robustness matters because uniformly high rater quality is rarely attainable in practice: large-scale crowdsourcing is inherently variable, screening is costly, and even diligent annotators are affected by subjective factors. BBQ is exactly the tool for this regime, extracting useful signal from partially reliable raters while suppressing the influence of noisy ones.

Table 1: Head-to-head comparison of the three BT-based aggregation methods across all eight datasets, with the best result in each column in **bold** and the second best underlined. Top: Top-1 agreement [%]; bottom: Kendall’s Tau; higher is better for both. BBQ attains the best top-1 agreement in every column and the best Kendall’s τ on five of eight datasets (second best on the other three). Its advantage is largest under noise: on IHQ-unscreened it reaches 61.92% top-1 agreement, roughly double Crowd-BT (33.15%) and Bayes-BT (24.32%).

	HUMAINE	MT-Bench	WD	HiFiC	ConHa	IHQ		
						all	scr.	unscr.
Top-1 Agreement [%]								
Crowd-BT	85.60	100.00	99.29	98.63	<u>66.59</u>	<u>85.31</u>	98.57	<u>33.15</u>
Bayes-BT	<u>97.30</u>	100.00	100.00	100.00	57.30	75.07	98.90	24.32
BBQ (ours)	97.50	100.00	100.00	100.00	77.12	99.32	99.80	61.92
Kendall’s Tau								
Crowd-BT	0.9385	0.9743	0.9279	<u>0.9366</u>	<u>0.9180</u>	<u>0.9245</u>	0.9171	0.8482
Bayes-BT	<u>0.9443</u>	0.9569	0.9459	0.9293	0.9110	0.9240	0.9116	<u>0.8507</u>
BBQ (ours)	0.9463	<u>0.9675</u>	<u>0.9359</u>	0.9525	0.9265	0.9270	<u>0.9132</u>	0.8563

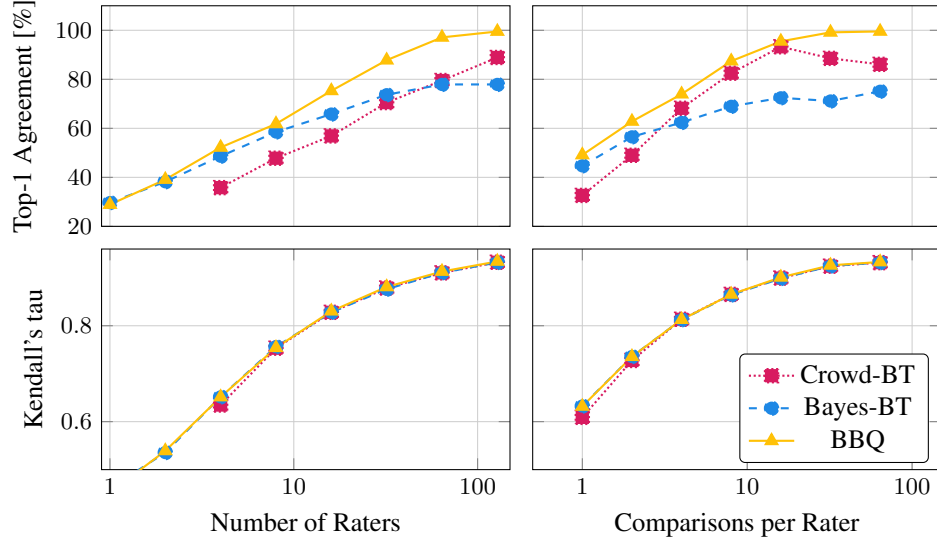


Figure 1: Scaling behavior of the three Bradley–Terry variants (Crowd-BT (Caron & Doucet, 2012), Bayes-BT (Chen et al., 2013), BBQ (ours)) on the IHQ-all dataset, with data added either by raters (**left**) or by comparisons per rater (**right**). Both top-1 agreement and Kendall’s τ improve monotonically with more data for every method. The two metrics separate the models differently: top-1 agreement clearly favors BBQ, which climbs to about 99.5% with 128 raters or 64 comparisons per rater, while Kendall’s τ stays close across methods. Two patterns stand out: Crowd-BT fails to converge with only one or two raters, exactly where BBQ’s EM updates remain stable, and once data is plentiful BBQ pulls ahead of Bayes-BT, whereas the two methods are comparable in the sparsest settings.

4.2 SCALING WITH RATERS AND COMPARISONS

Table 1 already showed that more comparisons per model tend to yield better performance; here we disentangle the two ways of adding comparisons, more raters or more comparisons per rater, and ask how each method responds.

Figure 1 isolates the two factors on the three BT variants: the left column varies the number of raters with comparisons per rater fixed at the maximum, and the right column varies comparisons per rater with the number of raters fixed at the maximum. The picture is consistent with the table:

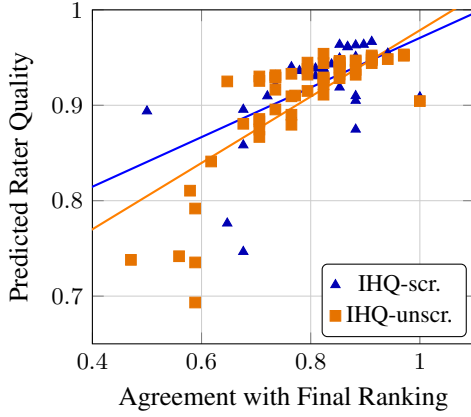


Figure 2: Predicted rater quality (y-axis) versus each rater’s agreement with the final ranking (x-axis) on the IHQ datasets, one point per rater (triangles: screened/filtered subset; squares: unscreened/unfiltered subset). The two are positively correlated in both subsets, and more strongly so once low-quality raters are included ($r = 0.72$ unfiltered vs. $r = 0.55$ filtered), confirming that BBQ’s estimated quality tracks how well a rater agrees with the consensus.

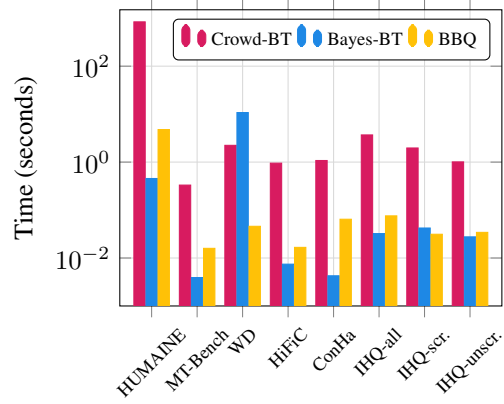


Figure 3: Average computation time per bootstrapped sample (seconds, log-scale) for the three methods across all eight datasets. BBQ stays within a few seconds on every dataset, whereas Crowd-BT is consistently slower and spikes to roughly 840 seconds (≈ 15 minutes) on HUMANE, up to three orders of magnitude slower than BBQ on the same data. Bayes-BT is competitive elsewhere but is the slowest on WD.

top-1 agreement clearly separates the methods in BBQ’s favor as data grows, while Kendall’s τ stays close across all three. The sharpest distinction is at the low-data end, where Crowd-BT fails to converge with only one or two raters, a direct manifestation of its lack of convergence guarantees, while BBQ’s EM updates remain stable throughout.

In the sparsest settings Crowd-BT and BBQ perform comparably, since BBQ needs both multiple raters and multiple comparisons per rater before it can tell rater qualities apart. Once data becomes plentiful, however, BBQ steadily pulls ahead of Bayes-BT along both axes. Bayes-BT is the weakest overall but occupies a useful niche: with very few raters or few comparisons per rater it can edge out Crowd-BT, precisely because Crowd-BT cannot reliably estimate rater quality when data is scarce.

4.3 RATER QUALITY

Figure 2 tests whether BBQ’s estimated rater quality is meaningful by plotting it against an independent quantity, each rater’s agreement with the final ranking. The relationship is clearly positive on both subsets: raters who agree more with the consensus are assigned higher quality. Quantitatively, the Pearson correlation is $r = 0.551$ over the 50 filtered raters and a stronger $r = 0.724$ over the 62 unfiltered raters, so the signal is strongest precisely when the population is noisiest and modeling quality matters most.

Consistent with this, the unscreened subset contains a visible cluster of raters with both low agreement and low predicted quality, and some fall below the 50% agreement level of random guessing, systematically disagreeing with the majority. BBQ flags exactly these raters with low quality scores, down-weighting their influence on the final ranking, which is the mechanism behind its robustness on the unscreened data reported above.

4.4 COMPUTATIONAL EFFICIENCY

Figure 3 reports the average wall-clock time each method needs to process a single bootstrapped sample, a practical proxy for feasibility when thousands of resamples are required. BBQ converges within a few seconds on all eight datasets without exception. Crowd-BT is slower than BBQ on

every dataset, and the gap becomes extreme on the largest one: on HUMAINE it takes roughly 15 minutes per sample, up to three orders of magnitude slower than BBQ. Bayes-BT is slightly faster than BBQ on most datasets but is the slowest of all three on WD, so BBQ is the only method that is fast everywhere. This uniform speed comes directly from the closed-form EM updates we derive, which sidestep the iterative gradient steps and learning-rate tuning that slow Crowd-BT down.

The comparison is if anything conservative: Crowd-BT is a heavily optimized C implementation (Kernighan & Ritchie, 1988), whereas BBQ runs in plain NumPy with no low-level optimization, so a compiled or vectorized version could widen BBQ’s speed advantage further. Taken together with the accuracy and stability results, this makes BBQ not just more robust but also the most practical choice for large-scale human preference studies and any workflow that relies on repeated bootstrapping.

4.5 CONFIDENCE INTERVALS

Beyond point rankings, we ask which methods report trustworthy uncertainty. In a controlled simulation under the null hypothesis H_0 that two items are equally skilled, each method estimates skills and builds uncertainty intervals, and we measure how often it wrongly declares a significant difference (Section G). The benchmark is exact: at the 99% confidence level a well-calibrated method should make this error 1% of the time. As shown in Figure 6, BBQ and Crowd-BT both hover just under this 1% target, so their intervals are well-calibrated, whereas Bayes-BT is markedly over-conservative at about 0.1%, roughly an order of magnitude below target, meaning its intervals are too wide to detect real differences. BBQ is thus the only method that is simultaneously well-calibrated here and best on ranking accuracy above.

5 LIMITATIONS

The clearest boundary of BBQ’s advantage is visible in our own results: on the high-quality, homogeneous datasets (MT-Bench, WD, IHQ-screened) it wins only by a fraction of a Kendall’s τ point, because when every rater is reliable there is little rater variability left to exploit. This is a narrowing of the margin, not a reversal, and it comes with a practical caveat, namely that guaranteeing uniformly high rater quality itself demands costly screening that is often infeasible at scale. The upshot is that BBQ delivers its largest gains exactly in the realistic, noisy crowdsourced regime, where the IHQ-unscreened results show it roughly doubling the top-1 agreement of the baselines, and its smallest gains where curation has already removed the noise it is designed to handle.

Another limitation concerns the type of data the framework can handle. BBQ is currently restricted to pairwise comparisons, whereas many human evaluation studies use ratings, rankings, or multi-way inputs. Extending the model to handle these forms of feedback would broaden its applicability.

Methodologically, our stability estimates come from a finite number of bootstrap resamples (10,000 per dataset, and 1,000 on HUMAINE for tractability); while this is enough to reveal the large and consistent margins reported above, still larger studies would further tighten the estimates. The model also assumes independence across comparisons and does not model contextual or order effects that can shape human judgments.

Finally, although BBQ already converges in seconds across all eight datasets, extremely large item or rater counts could still stress a plain NumPy implementation; as noted above, this is a headroom limit rather than a present bottleneck, since a compiled implementation would only extend the speed advantage we already observe.

6 CONCLUSION

We introduced BBQ, a Bayesian Bradley-Terry model that jointly estimates item quality and rater reliability. Our core contribution is the derivation of an Expectation-Maximization (EM) algorithm that simultaneously estimates item skills and individual rater quality, effectively down-weighting or removing the influence of unreliable participants. By explicitly modeling rater quality, the method produces more stable and accurate rankings. The EM algorithm ensures rapid convergence and monotonic likelihood improvement, addressing limitations of gradient-based approaches.

Across all eight datasets and against two competitive baselines, BBQ attained the best top-1 agreement in every case and the best Kendall’s τ on five of eight datasets, finishing a close second on the rest, a combination neither baseline matched. Its edge is largest exactly where evaluation is hardest: on unscreened crowdsourced comparisons it reached 61.92% top-1 agreement, roughly double the 33.15% and 24.32% of the baselines, and it retained near-best accuracy even when raters contributed fewer than 40 comparisons each. These gains did not come at the expense of speed or calibration: BBQ converged within seconds on every dataset, up to three orders of magnitude faster than Crowd-BT on the largest one, while its 99% intervals stayed close to the ideal 1% Type I error rate where the non-quality-aware Bayes-BT was over-conservative by an order of magnitude. Finally, the predicted rater-quality scores correlated with agreement to the final rankings (r up to 0.72), confirming that the model identifies reliable evaluators rather than merely fitting the data. Together these results show that incorporating rater quality matters most precisely when evaluation is heterogeneous or partially unreliable.

Overall, BBQ advances human evaluation methodology by offering a practical, interpretable, and generalizable approach to aggregate noisy pairwise comparisons. This work contributes a significant step toward more cost-effective, interpretable, and reproducible human studies for evaluating AI-generated content.

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A USAGE OF LLMs

During the preparation of this paper, we utilized large language models (LLMs) as supportive tools. ChatGPT, Claude, Gemini, and Grammarly assisted with spellchecking, refining phrasing, and condensing text to enhance clarity and readability. Additionally, ChatGPT, Claude, and Cursor were used for code analysis, completion, and the generation of visualizations to aid our development workflow. These tools were employed solely as auxiliary aids, while all core research ideas, experimental design, and interpretation of results were developed independently by the authors.

B BAYESIAN BT WITH RATER QUALITY DERIVATION

We consider R raters comparing K items. Each rater r has quality q_r , meaning that with probability q_r they follow the Bradley–Terry model, and with probability $1 - q_r$ they choose randomly. The following is a standard EM (Dempster et al., 1977) derivation that incorporates these rater-specific reliabilities into the Bayesian estimation framework.

B.1 MODEL DEFINITION

Notation:

- $\lambda = (\lambda_1, \dots, \lambda_K)$: skill parameters for the K items.
- $q = (q_1, \dots, q_R)$: quality parameters for the R raters.
- $Y_i \sim \mathcal{E}(\lambda_i)$: latent arrival time associated with item i .
- $A_{r,ij}^{(c)} \sim \text{Bernoulli}(q_r)$: indicator that the c -th comparison of pair (i, j) by rater r follows the Bradley–Terry model.
- $n_{r,ij}$: total number of comparisons of (i, j) by rater r .
- $n_{ij} = \sum_{r=1}^R n_{r,ij}$: total number of comparisons of (i, j) .
- $w_{r,ij}$: number of times rater r ranked i above j .
- $w_{ij} = \sum_{r=1}^R w_{r,ij}$: total number of times i was ranked above j .
- $m_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)}$: number of Bradley–Terry-model comparisons of (i, j) by rater r .
- $m_{ij} = \sum_{r=1}^R m_{r,ij}$: total number of Bradley–Terry-model comparisons of (i, j) .
- $v_{r,ij} = \sum_{c=1}^{w_{r,ij}} A_{r,ij}^{(c)}$: number of Bradley–Terry-model wins of i over j by rater r .
- $v_{ij} = \sum_{r=1}^R v_{r,ij}$: total number of Bradley–Terry-model wins of i over j .
- $Z_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)} \min(Y_i^{(c)}, Y_j^{(c)})$: sum of minimal arrival times, with

$$Z_{r,ij} \mid m_{r,ij} \sim \Gamma(m_{r,ij}, \lambda_i + \lambda_j).$$

Bradley–Terry probability:

$$P(i \text{ beats } j) = \frac{\lambda_i}{\lambda_i + \lambda_j}.$$

Mixture with rater quality:

$$P(r \text{ ranks } i \text{ above } j) = q_r \frac{\lambda_i}{\lambda_i + \lambda_j} + (1 - q_r) \frac{1}{2}.$$

Latent exponential view:

$$Y_i \sim \mathcal{E}(\lambda_i), \quad Y_j \sim \mathcal{E}(\lambda_j), \quad P(i \text{ beats } j) = P(Y_i < Y_j).$$

B.2 COMPLETE-DATA LOG-LIKELIHOOD

We need to compute:

$$\begin{aligned}\ell_c(\lambda, q; D, Z, A) &= \log P(D, Z, A \mid \lambda, q) \\ &= \log P(Z \mid D, A, \lambda, q) + \log P(A \mid D, \lambda, q) + \log P(D \mid \lambda, q)\end{aligned}$$

Log-Likelihood of $P(Z \mid A, \lambda)$:

$$\begin{aligned}\log P(Z \mid D, A, \lambda, q) &= \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{ij} \log(\lambda_i + \lambda_j) - (\lambda_i + \lambda_j) z_{ij} \right. \\ &\quad \left. + (m_{ij} - 1) \log z_{ij} - \log \Gamma(m_{ij}) \right]\end{aligned}$$

Log-Likelihood of $P(A \mid q)$:

$$\log P(A \mid D, \lambda, q) = \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{r,ij} \log q_r + (n_{r,ij} - m_{r,ij}) \log(1 - q_r) \right]$$

Log-Likelihood of $P(D \mid \lambda, A)$:

$$\begin{aligned}\log P(D \mid \lambda, q) &= \sum_{r=1}^R \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K \left[v_{r,ij} \log \frac{\lambda_i}{\lambda_i + \lambda_j} + (w_{r,ij} - v_{r,ij}) \log \frac{1}{2} \right] \\ &= \sum_{r=1}^R \sum_{i=1}^K [v_{ir} \log \lambda_i] \\ &\quad - \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K [(v_{r,ij} + v_{r,ji}) \log(\lambda_i + \lambda_j) + (w_{r,ij} + w_{r,ji} - v_{r,ij} - v_{r,ji}) \log 2] \\ &= \sum_{i=1}^K [v_i \log \lambda_i] \\ &\quad - \sum_{i=1}^K \sum_{j=i+1}^K [m_{ij} \log(\lambda_i + \lambda_j) + (n_{ij} - m_{ij}) \log 2]\end{aligned}$$

Complete-Data Log-Likelihood:

$$\begin{aligned}
\ell_c(\lambda, q; D, Z, A) &= \log P(Z \mid D, A, \lambda, q) + \log P(A \mid D, \lambda, q) + \log P(D \mid \lambda, q) \\
&= \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{ij} \log(\lambda_i + \lambda_j) - (\lambda_i + \lambda_j) z_{ij} + (m_{ij} - 1) \log z_{ij} - \log \Gamma(m_{ij}) \right] \\
&\quad + \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{r,ij} \log q_r + (n_{r,ij} - m_{r,ij}) \log(1 - q_r) \right] \\
&\quad + \sum_{i=1}^K [v_i \log \lambda_i] \\
&\quad - \sum_{i=1}^K \sum_{j=i+1}^K [m_{ij} \log(\lambda_i + \lambda_j) + (n_{ij} - m_{ij}) \log 2] \\
&= \sum_{i=1}^K \sum_{j=i+1}^K \left[(n_{ij} - m_{ij}) \log 2 - (\lambda_i + \lambda_j) z_{ij} + (m_{ij} - 1) \log z_{ij} - \log \Gamma(m_{ij}) \right] \\
&\quad + \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{r,ij} \log q_r + (n_{r,ij} - m_{r,ij}) \log(1 - q_r) \right] \\
&\quad + \sum_{i=1}^K [v_i \log \lambda_i]
\end{aligned}$$

B.3 EXPECTATION STEP

We introduce conjugate priors: Gamma distribution $\lambda_i \sim \Gamma(a, b)$ for each item i , and Beta $q_r \sim B(\alpha, \beta)$ for each rater r .

The Q -function is the expectation of the complete-data log-posterior:

$$Q(\lambda, q \mid \lambda^*, q^*) = \mathbb{E}_{A, Z \mid D, \lambda^*, q^*} [\ell_c(\lambda, q; D, Z, A) + \log P(\lambda) + \log P(q)].$$

Posterior probability of a Bradley-Terry-consistent annotation:

$$\begin{aligned}
P(A_{r,ij}^{(k)} = 1 \mid i \succ j) &= \frac{P(i \succ j \mid A = 1) P(A = 1)}{P(i \succ j \mid A = 1) P(A = 1) + P(i \succ j \mid A = 0) P(A = 0)} \\
&= \frac{\left(\frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*} \right) q_r^*}{\left(\frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*} \right) q_r^* + \frac{1}{2} (1 - q_r^*)} = \gamma_{r,ij}^*.
\end{aligned}$$

Expected sufficient statistics:

$$\begin{aligned}
\mathbb{E}[m_{r,ij}] &= w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^* \\
\mathbb{E}[m_{ij}] &= \sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \\
\mathbb{E}[v_{r,ij}] &= w_{r,ij} \gamma_{r,ij}^* \\
\mathbb{E}[v_{ir}] &= \sum_{j:j \neq i}^K w_{r,ij} \gamma_{r,ij}^* \\
\mathbb{E}[v_i] &= \sum_{r=1}^R \sum_{j:j \neq i}^K w_{r,ij} \gamma_{r,ij}^* \\
\mathbb{E}[z_{ij}] &= \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*}.
\end{aligned}$$

Expected complete-data log-posterior:

$$\begin{aligned}\mathbb{E}[\ell_c] = & \sum_{i=1}^K \sum_{j=i+1}^K \left[-(\lambda_i + \lambda_j) \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} \right] \\ & + \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[(w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \log q_r \right. \\ & \quad \left. + (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) \log(1 - q_r) \right] \\ & + \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K w_{r,ij} \gamma_{r,ij}^* \log \lambda_i \\ & + \text{const.}\end{aligned}$$

Priors contribute:

$$\begin{aligned}\mathbb{E}[\log P(\lambda)] &= \sum_{i=1}^K [(a-1) \log \lambda_i - b \lambda_i], \\ \mathbb{E}[\log P(q)] &= \sum_{r=1}^R [(\alpha-1) \log q_r + (\beta-1) \log(1 - q_r)].\end{aligned}$$

Final Q -function:

$$\begin{aligned}Q(\lambda, q \mid \lambda^*, q^*) = & \sum_{i=1}^K \sum_{j=i+1}^K \left[-(\lambda_i + \lambda_j) \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} \right] \\ & + \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[(w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \log q_r \right. \\ & \quad \left. + (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) \log(1 - q_r) \right] \\ & + \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K w_{r,ij} \gamma_{r,ij}^* \log \lambda_i \\ & + \sum_{i=1}^K [(a-1) \log \lambda_i - b \lambda_i] \\ & + \sum_{r=1}^R [(\alpha-1) \log q_r + (\beta-1) \log(1 - q_r)] \\ & + \text{const.}\end{aligned}$$

where

$$\gamma_{r,ij}^* = \frac{q_r^* \frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*}}{q_r^* \frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*} + (1 - q_r^*) \frac{1}{2}}.$$

B.4 M-STEP

The M-step maximizes the Q -function w.r.t. the parameters (λ, q) , holding the expectations computed in the E-step fixed.

Update for q_r The update for each rater quality q_r is obtained by maximizing Q with respect to q_r (including the Beta prior).

$$\begin{aligned}
Q(q \mid \lambda^*, q^*) &= \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[(w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \log q_r \right. \\
&\quad \left. + (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) \log(1 - q_r) \right] \\
&\quad + (\alpha - 1) \log q_r + (\beta - 1) \log(1 - q_r) + \text{const.} \\
\frac{\partial Q}{\partial q_r} &= \sum_{i=1}^K \sum_{j=i+1}^K \left[\frac{w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*}{q_r} - \frac{n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{1 - q_r} \right] \\
&\quad + \frac{\alpha - 1}{q_r} - \frac{\beta - 1}{1 - q_r} \\
&\stackrel{0}{=} \frac{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) + (\alpha - 1)}{q_r} \\
&\quad - \frac{\sum_{i=1}^K \sum_{j=i+1}^K (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) + (\beta - 1)}{1 - q_r} \\
&\Rightarrow (1 - q_r) \left(\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) + (\alpha - 1) \right) \\
&\Rightarrow q_r \left(\sum_{i=1}^K \sum_{j=i+1}^K (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) + (\beta - 1) \right) \\
&\Rightarrow q_r = \frac{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) + (\alpha - 1)}{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^* + n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) + (\beta + \alpha - 2)} \\
&\Rightarrow q_r^{(t)} = \frac{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)}) + (\alpha - 1)}{n_r + \beta + \alpha - 2}
\end{aligned}$$

where the last line gives the explicit update at iteration t .

Update for λ_i The update for each item skill λ_i is obtained by maximizing Q with respect to λ_i (including the Gamma prior).

$$\begin{aligned}
Q(\lambda \mid \lambda^*, q^*) &= \sum_{i=1}^K \sum_{j=i+1}^K -(\lambda_i + \lambda_j) \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} \\
&\quad + \sum_{r=1}^R \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K w_{r,ij} \gamma_{r,ij}^* \log \lambda_i \\
&\quad + \sum_{i=1}^K [(a - 1) \log \lambda_i - b \lambda_i] + \text{const.}
\end{aligned}$$

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$$\begin{aligned}
\frac{\partial Q}{\partial \lambda_i} &= - \sum_{j \neq i}^K \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} + \sum_{r=1}^R \frac{\sum_{j \neq i}^K w_{r,ij} \gamma_{r,ij}^*}{\lambda_i} + \frac{a-1}{\lambda_i} - b \\
0 &\stackrel{!}{=} \frac{\sum_{r=1}^R \sum_{j \neq i}^K w_{r,ij} \gamma_{r,ij}^* + (a-1)}{\lambda_i} - \sum_{j \neq i}^K \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} - b \\
\Rightarrow \lambda_i^{(t)} &= \frac{\sum_{r=1}^R \sum_{j \neq i}^K w_{r,ij} \gamma_{r,ij}^{(t-1)} + (a-1)}{\sum_{j \neq i}^K \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)})}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}} + b} \\
\gamma_{r,ij}^{(t-1)} &= \frac{q_r^{(t-1)} \frac{\lambda_i^{(t-1)}}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}}}{q_r^{(t-1)} \frac{\lambda_i^{(t-1)}}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}} + (1 - q_r^{(t-1)})^{\frac{1}{2}}}
\end{aligned}$$

where the last line gives the explicit update at iteration t .

C HYPERPARAMETERS

For Crowd-BT, we use the implementation of Google (2025) with the default hyperparameters.

The only hyperparameters in the Bayes-BT and BBQ models are the prior distribution parameters and the stopping thresholds. For both Bayes-BT and BBQ, we consider the model converged when no ELO score changes by more than 1 between two iterations. The ELO score can be calculated from the skill parameter λ as:

$$\text{ELO} = \log(\text{skill}) \cdot \text{ELO_SCALE_FACTOR}. \quad (14)$$

where we set the ELO.SCALE.FACTOR to 400.

We chose a gamma prior with shape 5 and rate 0.1 for the skill parameters in both Bayes-BT and BBQ. For BBQ, the beta prior on the rater quality has $\alpha = 10$ and $\beta = 2$.

D DATASETS

We evaluate our models on a diverse set of human preference datasets covering both language and image domains. Table 2 provides an overview.

For natural language, we use HUMAINE (Team, 2025), a large-scale benchmark with 105,220 comparisons from 1,977 raters, and MT-Bench (Zheng et al., 2023), with 3,355 crowd-worker judgments on multi-turn dialogues.

For image compression, we use three datasets that deliberately span the expertise spectrum: WD (Ballé et al., 2025), a dense expert-labeled set with about 3,332 comparisons per rater from only 5 raters; HiFiC (Mentzer et al., 2020), which evaluates learned image codecs at a moderate 261 comparisons per rater; and ConHa (Aczel & Wattenhofer, 2024), a smaller conditional-generation study at about 38 comparisons per rater. This range in per-rater density is exactly the axis along which the methods differ most, making the set a demanding testbed for robustness.

Finally, we introduce the inhomogeneous rater quality (**IHQ**) dataset, obtained from a user study on the CLIC2024 (CLIC, 2025) data conducted via the *Mabyduck* platform. It contains two-alternative-forced-choice (2AFC) judgments across 28 generative image compression models. To study the impact of rater quality, we provide two subsets: (i) **screened**, where raters passed pre-screening checks for attention and display quality, and (ii) **unscreened**, where all raters are included. The screened subset represents higher-quality raters, whereas the unscreened subset better reflects the noisy conditions typical of large-scale human evaluations.

Table 2: Summary of datasets used in our experiments. The IHQ dataset was collected by us on the Mabyduck platform and includes both 2AFC and 3AFC settings. For this dataset, we provide screened and unscreened subsets: screened subsets include only raters who passed pre-screening tests for attention and display quality, ensuring higher reliability, while unscreened subsets include all raters.

dataset	# comparisons	# raters	# models	comp/rater	comp/model
HUMAINE (Team, 2025)	105,220	1,977	27	53.2	3897.0
MT-Bench (Zheng et al., 2023)	3,355	65	6	51.6	559.2
WD (Ballé et al., 2025)	16,659	5	30	3331.8	555.3
HiFiC (Mentzer et al., 2020)	5,220	20	9	261.0	580.0
ConHa (Aczel & Wattenhofer, 2024)	1,531	40	8	38.3	191.4
IHQ-all	4,074	112	28	36.4	145.5
IHQ-screened	2,012	50	28	40.2	71.9
IHQ-unscreened	2,062	62	28	33.3	73.6

Together these eight datasets span a wide operating range, from HUMAINE’s 105,220 comparisons down to ConHa’s 1,531 and from 5 expert raters (WD) to nearly 2,000 (HUMAINE), letting us compare the methods on both large, relatively clean benchmarks and small, noisy settings where modeling rater reliability is critical.

E USER STUDY PLATFORM

All pairwise comparisons on the CLIC2024 (CLIC, 2025) dataset were collected using the Mabyduck platform (Ltd., 2025). A screenshot of the platform can be seen in Figure 4. The task asks: “Which image looks more similar to the reference image?” Ties are not allowed, and all pairs were selected uniformly at random.

To study rater quality, we split the IHQ dataset into screened and unscreened subsets. Screened raters passed pre-tests for color vision, display contrast, and sensitivity to subtle differences, and completed training comparisons choosing the higher-quality image to reinforce evaluation criteria.

Figure 5 shows four example images from the pre-screening process. The first two are standard color blindness tests, where raters must correctly identify the number shown in each pattern. The final two images are shape detection tests designed to evaluate the raters’ sensitivity to low-contrast objects: one features a light gray shape on a white background, and the other a dark gray shape on a black background. These pre-screening tests help ensure that only raters with adequate visual capabilities contribute to the screened subset.

F BOOTSTRAPPING DETAILS

Because both BBQ and Crowd-BT explicitly model rater quality, we bootstrap over raters rather than over individual comparisons; this preserves each rater’s comparison distribution and gives all three methods a fair, like-for-like assessment of stability. We use 10,000 resamples for every dataset, with a single exception dictated purely by baseline runtime: on HUMAINE a single Crowd-BT iteration takes about 15 minutes (Section 4.4), so we reduce it to 1,000 resamples there. Notably, this bottleneck stems from the Crowd-BT baseline rather than BBQ, which processes the same data in seconds.

G UNCERTAINTY ESTIMATION DETAILS

In addition to ranking items, estimating uncertainty is important to assess whether observed differences are statistically significant.

The Bayesian Bradley-Terry (Bayes-BT) and Bayesian Bradley-Terry with Quality (BBQ) models quantify uncertainty via the posterior distribution over item skills. Each skill has a Gamma prior,

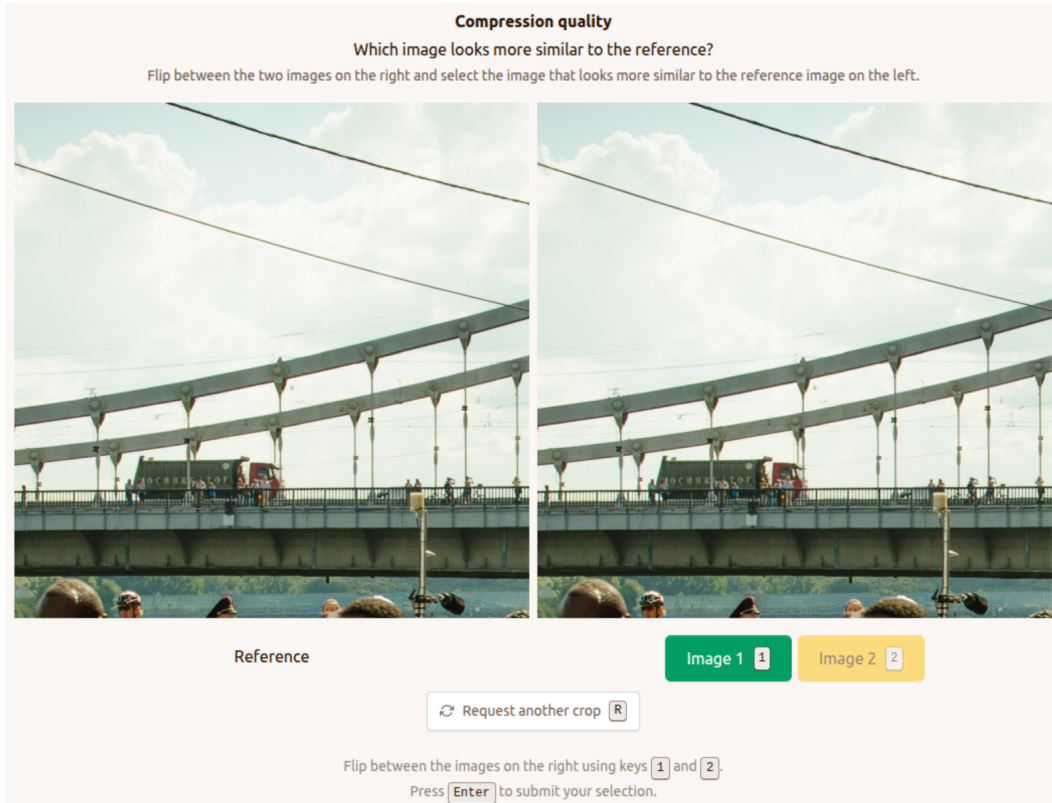


Figure 4: Screenshot of the Mabyduck user study platform used for collecting pairwise comparisons. A reference image is shown on the left, and the rater selects between two compressed images on the right.

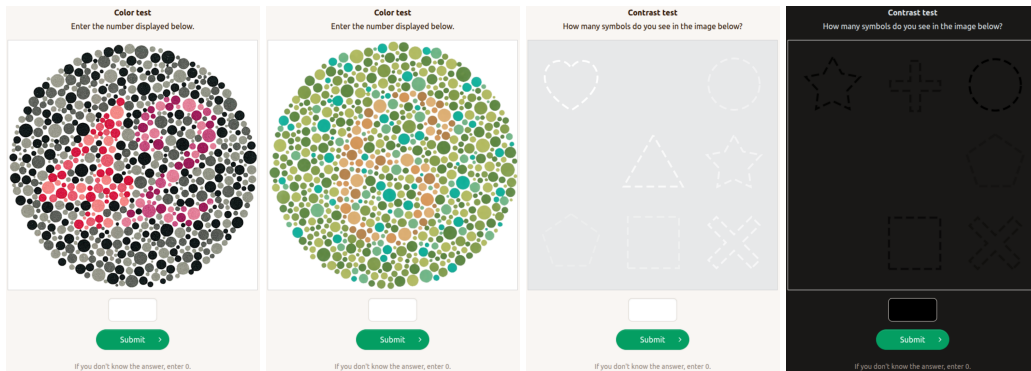


Figure 5: Four example images from the pre-screening process for raters. The first two are color blindness tests, where raters must identify the number displayed in each pattern. The last two are shape detection tests designed to evaluate sensitivity to low-contrast objects: one light gray shape on a white background and one dark gray shape on a black background.

which is updated using the observed pairwise comparisons. Credible intervals derived from the posterior are then converted to the Elo scale to facilitate comparison across items.

The classical Crowd-BT model, being non-Bayesian, estimates uncertainty using a frequentist approximation. Specifically, a second-order Taylor expansion around the maximum likelihood estimate is employed. The Hessian of the log-likelihood is inverted to obtain the covariance matrix, whose diagonal entries correspond to the variances of individual items. These variances are con-

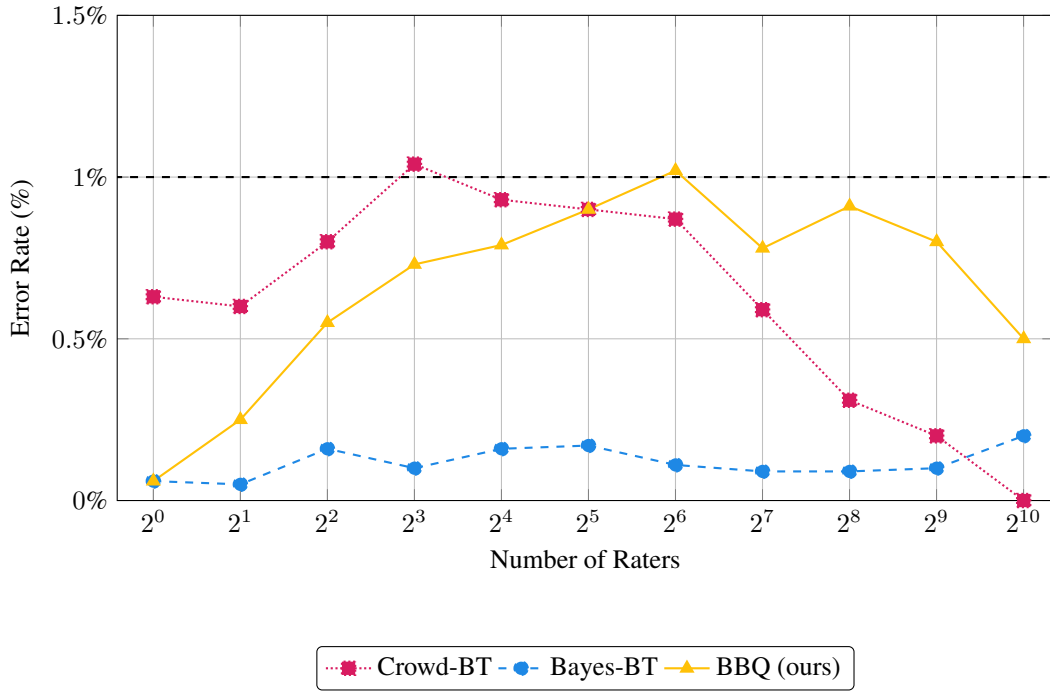


Figure 6: Type I error rate versus the number of raters under H_0 (equally skilled items); the dashed line marks the ideal 1% rate at the 99% confidence level. BBQ and Crowd-BT track the 1% target closely across rater counts, indicating well-calibrated intervals, whereas Bayes-BT sits far below it (around 0.1%), revealing overly conservative intervals.

verted to 99% confidence intervals via:

$$p_{99} = \sqrt{\text{diag}(\text{covariance})} \times k_{\text{Erfc}0.01} \times \sqrt{2} \approx \sqrt{\text{diag}(\text{covariance})} \times 3.29,$$

where the constants scale the standard deviation to match the 99% confidence level. Narrower likelihood peaks yield smaller intervals, reflecting higher certainty, while flatter peaks produce larger intervals, indicating greater uncertainty.

To evaluate how well these models estimate uncertainty, we conduct a controlled simulation experiment. We generate trials with two items of equal skill (50–50 win probability) and simulate pairwise comparison data using coin flips. For each trial, multiple raters (users) are simulated, and models are applied to estimate Elo scores and their associated uncertainty.

Formally, the null hypothesis H_0 states that the two items are equally strong. For each trial, we compute the 99% confidence (or credible) interval for each item’s skill estimate. A model is said to incorrectly reject H_0 if the confidence intervals do not overlap, indicating a statistically significant difference between items when none exists. The primary metric of interest is the frequency with which each model incorrectly concludes that the items are different, i.e., the observed Type I error rate.

We run 10,000 trials with 50 comparisons per rater, sweeping the number of raters to see how calibration scales with data, and compare each model’s empirical Type I error rate against the theoretical 1% expected at the 99% confidence level. As shown in Figure 6, BBQ and Crowd-BT stay close to the 1% target across the full sweep, so both are well-calibrated for significance testing. Bayes-BT, in contrast, is consistently and substantially below target (around 0.1%), meaning its credible intervals are too wide and it forfeits statistical power. BBQ therefore matches the calibration of Crowd-BT while, as Section 4 shows, being both faster and more accurate on ranking.